A shared Pell certificate for Con(ZFC)

An explicit quartic in 1,873 unrestricted integer variables

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with ChatGPT assistance

23 October 2026

Repository baseline: `b594a5e988e986f695923d068ef3d33ab2ada16f`

Abstract

We give a completely materialized integer polynomial P of degree four, in 1,873 unrestricted integer variables, for which an ordinary proof formalizable in ZFC establishes

$$\mathrm{ZFC} \vdash \left[\forall \mathbf{z} \in \mathbb{Z}^{1873}\ P(\mathbf{z}) \neq 0\right] \longleftrightarrow \mathrm{Con(ZFC)}.$$

The polynomial is a sum of 1,468 explicitly supplied quadratic squares; its collected expansion has 77,523 nonzero monomials and coefficient height $1{,}323{,}565{,}948{,}780{,}873{,}728 < 2^{61}$. The complete coefficients, deterministic generator, independent exact checkers, source interfaces and proof notes accompany this article.

The main new result is a simultaneous Pell certificate: forty positive power graphs share one bootstrap, one auxiliary Pell solution and one CRT base. Equal exponents share coordinates; three nearby exponents are recovered by recurrences. A signed index recovery lemma removes both auxiliary sign guards for each of sixteen seed indices. A single common output bound replaces forty private bounds. These are existence-preserving mathematical transformations with explicit completeness constructions, rather than informal identification of unrelated existential witnesses.

The fixed logical input is the repository's C222 numerical calculus. Its established ZFC-refutation calibration is inherited. No K215 simulation or Jones universal polynomial is used. Relative to the latest incoming direct quartic, the integer-variable count falls from 7,617 to 1,873. This is a reduction within the direct calculus construction; it is not a claim of a globally smallest consistency equation or a new minimum among all quartics. No new Lean or encoded object-language ZFC proof certificate is claimed.

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1 The result and its exact scope

The user-facing equation is the following ordinary polynomial equation:

$$\boxed{P(z_0,\dots,z_{1872}) = \sum_{\nu=0}^{1467} Q_\nu(z_0,\dots,z_{1872})^2 = 0, \qquad (z_0,\dots,z_{1872}) \in \mathbb{Z}^{1873}.} \tag{1}$$

Every Q_ν is explicitly listed in `equations/residuals.json`. The collected integer coefficients of P itself are explicitly listed in `equations/polynomial.json`. Neither file contains a factorial, variable exponent, binomial operator, bounded quantifier, unspecified Diophantine representation, external constant port, or oracle call. The files are two exact representations of the same finite polynomial.

Theorem 1.1 (Fixed consistency equation)**.** *For the polynomial specified by* (1) *and the accompanying coefficient list, ZFC proves*

$$(\exists \mathbf{z} \in \mathbb{Z}^{1873}\ P(\mathbf{z}) = 0) \quad \longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}\,.$$

Here $\mathrm{Ref}_{\mathrm{ZFC}}$ *means existence of a finite syntactic derivation of falsity in the repository's stated ZFC proof system. Consequently ZFC proves the no-zero/*$\mathrm{Con}(\mathrm{ZFC})$ *biconditional.*

The proof composes the inherited C222 calibration with the new arithmetic transformations proved below. It does not assume consistency, the existence of a standard model of ZFC, or a bound on a possible refutation. The output is a mathematical proof with exact finite computational certificates. The complete composed theorem has not been checked by a proof assistant during this work.

1.1 Two supplied quartics

Measure	Primary	Smaller coefficient/term variant
Integer variables	1,873	3,292
Degree	4	4
Quadratic residuals	1,468	2,887
Collected monomials	77,523	11,218
Coefficient height	1,323,565,948,780,873,728	14,869,246
Coefficient bit length	61	24

The second equation is in `equations_small_height/`. It keeps more affine arithmetic registers, so its individual residuals have at most five monomials. Both variants have the same semantic certificate and the same consistency theorem. Their differences are exact choices of intermediate registers, with unique integral restoration.

1.2 Comparison with the inspected repository

The comparison baseline is the actual repository at the commit printed on the title page, including the then-new `ConZFC_Direct_Projection_7617.zip`, rather than only the older manuscript-collection README [1, 4, 2].

Direct integer construction	Variables	Degree	Monomials
Earlier direct C222 round 12	27,916	4	758,275
Shared CRT, prime-power lift	8,619	4	39,811
Incoming eight-column projection	7,617	4	148,220
This work, primary	**1,873**	4	77,523
This work, retained affine registers	3,292	4	**11,218**

The primary count is about 75.4% below 7,617 and 78.3% below 8,619. The latter parent has fewer monomials than the primary; the second supplied variant addresses that tradeoff. These comparisons are not uniform claims of improvement in every measure.

The repository also records nine-positive-variable and ten-integer-variable high-degree equations, and quartics with 37 integer variables using the native coefficient construction [3]. Thus 1,873 is not a record variable count among all available routes. The contribution here is a substantially smaller direct proof-calculus equation with fully expanded, moderate-size literal coefficients and a reusable simultaneous arithmetic theorem. Historical priority for the underlying idea is not asserted.

2 Pell notation and the inherited finite theorems

For $a \geq 2$ define

$$\begin{aligned} \chi_a(0) &= 1, & \psi_a(0) &= 0, \\ \chi_a(j+1) &= a\chi_a(j) + (a^2-1)\psi_a(j), & \psi_a(j+1) &= \chi_a(j) + a\psi_a(j). \end{aligned}$$

Equivalently, $(a+\sqrt{a^2-1})^j = \chi_a(j) + \psi_a(j)\sqrt{a^2-1}$. All quantities used in the proof can instead be defined by the displayed integer recurrence. In particular

$$\chi_a(j)^2 - (a^2-1)\psi_a(j)^2 = 1.$$

For $j > 0$, both coordinates are positive, $\psi_a(j) \geq j$, and $\chi_a(j) \geq a^j$. Every positive-abscissa nonnegative-ordinate solution of this Pell equation is one of these pairs.

The inherited arithmetic input is explicit: the actual statements and proofs of `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in Mathlib [7], together with the repository's literal polynomial interfaces `PellRepresentation.lean` and `PowerRepresentation.lean`. Copies are included in `provenance/`. The inspected Mathlib blob is `db9c757dd7cac29260397150fd190d72715ea3f8`.

Proposition 2.1 (Positive power criterion)**.** *For $n, k > 0$ and $m \geq 0$, $m = n^k$ is equivalent to the existence of natural w, a, t, z with $a \geq 2$ satisfying*

$$\begin{aligned} \chi_a(k) &\equiv \psi_a(k)(a-n) + m \pmod{t}, \qquad 2an = t + n^2 + 1, \\ m &< t, \qquad n \leq w, \quad k \leq w, \qquad a^2 - ((w+1)^2 - 1)(wz)^2 = 1. \end{aligned}$$

In the formal source $a - n$ is natural subtraction. Our construction proves $a > n$ before applying this proposition.

The usual finite index certificate for $(x, y) = (\chi_a(k), \psi_a(k))$ uses three Pell norms and four congruences. The next section proves the signed variant needed here, with all domain conditions stated explicitly. This removes a genuine arithmetic restriction from the compiler; it is not just a relabeling of variables.

3 Signed recovery of a positive Pell index

We use two precise properties of Pell sequences [7]:

1. If $\psi_a(i)^2 \mid \psi_a(n)$, then $\psi_a(i) \mid n$. This is `dvd_of_ysq_dvd` in the supplied source.
2. If $0 < i \leq n$ and $\chi_a(h) \equiv \chi_a(i) \pmod{\chi_a(n)}$, then $h \equiv i$ or $h \equiv -i \pmod{4n}$. This is `modEq_of_xn_modEq`; its positivity hypothesis excludes the exceptional zero-index coincidence in the preceding auxiliary lemmas.

The addition formula also gives

$$\chi_a(h+2n) \equiv -\chi_a(h) \pmod{\chi_a(n)}. \tag{2}$$

Indeed $\chi_a(2n) = 2\chi_a(n)^2 - 1$ and $\psi_a(2n) = 2\chi_a(n)\psi_a(n)$. These facts specify exactly what is needed from the modular Pell theory; no empirical pattern is substituted for an index theorem.

Lemma 3.1 (Signed index recovery)**.** *Let $A, B, U, X, Y, k \in \mathbb{N}$, with $A, B \geq 2$, $U \geq 2$, $X > 0$ and $1 \leq k \leq Y$. Let $V, S, R \in \mathbb{Z}$. Assume*

$$X^2 - (A^2-1)Y^2 = 1, \quad U^2 - (A^2-1)V^2 = 1, \quad S^2 - (B^2-1)R^2 = 1, \tag{3}$$

$$Y^2 \mid V, \quad B \equiv A \pmod{U}, \quad B \equiv 1 \pmod{4Y}, \quad S \equiv X \pmod{U}, \quad R \equiv k \pmod{4Y}. \tag{4}$$

Then $X = \chi_A(k)$ and $Y = \psi_A(k)$. No sign hypothesis on S or R is needed.

Proof. Since $U \geq 2$, the second norm forces $V \neq 0$. Pell classification gives indices i, n, h with

$$(X, Y) = (\chi_A(i), \psi_A(i)), \quad (U, |V|) = (\chi_A(n), \psi_A(n)), \quad (|S|, |R|) = (\chi_B(h), \psi_B(h)).$$

We have $i, n > 0$. From $Y^2 \mid |V|$ we obtain $Y \leq |V|$, hence $i \leq n$ by strict monotonicity; also $Y \mid n$ by property 1 above.

The coordinate polynomials preserve congruences in their base. Consequently $B \equiv A \pmod{U}$ and the congruence for S imply

$$\chi_A(h) \equiv \pm\chi_A(i) \pmod{U}.$$

In the positive-sign case property 2 gives $h \equiv \pm i \pmod{4n}$. In the negative-sign case apply (2) and property 2 to $h + 2n$. In both cases

$$h \equiv \pm i \pmod{2n}. \tag{5}$$

This step is why taking absolute values alone would have been insufficient: the possible change of sign is absorbed in the index modulus.

At base 1 the recurrence has ordinate h. Thus $B \equiv 1 \pmod{4Y}$ implies $\psi_B(h) \equiv h \pmod{4Y}$. The signed congruence for R therefore gives

$$h \equiv \pm k \pmod{4Y}.$$

Reduce this congruence and (5) modulo $2Y$, which divides $2n$. It follows that $i \equiv \pm k \pmod{2Y}$. Finally $1 \leq i \leq Y$ because $\psi_A(i) \geq i$, and $1 \leq k \leq Y$ by hypothesis. The difference congruence gives $i = k$; the sum congruence can only give $i + k = 2Y$, which also forces $i = k = Y$. This includes the boundary cases $Y = 1$ and $i = n$. □

In the final polynomial, this lemma permits the two variables in $S = X + Us$ and $R = k + 4Yr$ to range over all integers. For sixteen seed indices it removes 32 nonnegativity equations and their 96 auxiliary integer coordinates. All remaining necessary sign restrictions are kept.

4 One simultaneous certificate for all powers

Fix a nonempty finite list of positive input/output triples (n_i, k_i, m_i), $1 \le i \le q$. Initially group identical exponents into r classes, with exponent e_j and class map $j(i)$. Introduce natural variables g, u, b, x_j, y_j and arbitrary integer variables $z, v, \lambda_B, s_j, r_j, \lambda_i$. Define polynomial abbreviations

$$W = \sum_{i=1}^{q}(n_i + k_i), \qquad A = 2 + \sum_{i=1}^{q} m_i + g, \qquad U = u + 2, \tag{6}$$

$$X_j = x_j + 1, \qquad Y_j = e_j + y_j, \qquad P_* = \prod_{j=1}^{r} Y_j, \tag{7}$$

$$V = 4P_*^2 v, \qquad B = 1 + 4P_*(b+1), \qquad T_i = 2An_i - n_i^2 - 1, \tag{8}$$

$$S_j = X_j + Us_j, \qquad R_j = e_j + 4Y_j r_j. \tag{9}$$

The star distinguishes P_* from the final quartic P. These abbreviations introduce no quantified variables.

Impose

$$A^2 - ((W+1)^2 - 1)(Wz)^2 = 1, \tag{10}$$

$$U^2 - (A^2 - 1)V^2 = 1, \tag{11}$$

$$B - A = U\lambda_B, \tag{12}$$

$$X_j^2 - (A^2 - 1)Y_j^2 = 1 \qquad (1 \le j \le r), \tag{13}$$

$$S_j^2 - (B^2 - 1)R_j^2 = 1 \qquad (1 \le j \le r), \tag{14}$$

$$X_{j(i)} + Y_{j(i)}n_i - Y_{j(i)}A - m_i = T_i\lambda_i \qquad (1 \le i \le q). \tag{15}$$

An emitted congruence may use the negative of the displayed quotient; since these quotients are unrestricted integers this is an exact coordinate change.

Theorem 4.1 (Simultaneous power graph)**.** *For specified positive triples (n_i, k_i, m_i), all equalities $m_i = n_i^{k_i}$ hold if and only if (10)–(15) have auxiliary values in the stated domains.*

4.1 Soundness and the single common output bound

First $W \ge 2$, $A \ge 2$, $U \ge 2$, $Y_j \ge e_j \ge 1$, $P_* \ge 1$, and $B \ge 5$. If $z = 0$, (10) would give $A^2 = 1$. Therefore $z^2 \ge 1$, and rearrangement gives

$$A^2 = 1 + W^3(W+2)z^2 \ge 1 + W(W+2) = (W+1)^2.$$

Thus $A \ge W + 1 > n_i$ for every call. The exact identity

$$T_i - A = (n_i - 1)(2A - n_i - 1) + (A - 2) \tag{16}$$

shows $T_i \ge A > m_i$. This argument uses only the displayed polynomial system and the required domains. In particular it does not first assume that the candidate power outputs are correct. One global gap g has replaced all q private strict output-bound slacks.

Equation (11) forces $v \ne 0$. Since each Y_j divides P_*, we have $Y_j^2 \mid V$ and $B \equiv 1 \pmod{4Y_j}$. The definitions of S_j, R_j and (12) give the remaining congruences in Lemma 3.1. That lemma identifies every (X_j, Y_j) with the true pair at e_j in base A.

Now (15), (16), $n_i, k_i \le W$, and (10) with $|z|$ satisfy Proposition 2.1. The natural subtraction there agrees with integer subtraction because $A > n_i$. Hence every $m_i = n_i^{k_i}$.

4.2 Completeness without incompatible shared choices

Suppose all candidate outputs are correct. Fix all input/output values first. This is legitimate even if the surrounding arithmetic system relates different calls; a simultaneous true tuple is already given.

Stage 1: the common bootstrap. At base $W+1$ the recurrence gives $\psi_{W+1}(\ell) \equiv \ell \pmod{W}$. Choose a sufficiently large positive multiple $\ell = Wh$ and set

$$A = \chi_{W+1}(\ell) \geq 2 + \sum_i m_i, \qquad z = \psi_{W+1}(\ell)/W, \qquad g = A - 2 - \sum_i m_i.$$

The sequence is increasing and unbounded, so such an h exists. All choices are finite natural-number constructions, and (10) holds. Define $X_j = \chi_A(e_j)$ and $Y_j = \psi_A(e_j)$. Then $x_j = X_j - 1$ and $y_j = Y_j - e_j$ are nonnegative.

Stage 2: the common auxiliary Pell solution. Form $P_* \geq 1$ and $K = 4P_*^2$. The matrix

$$M_A = \begin{pmatrix} A & A^2 - 1 \\ 1 & A \end{pmatrix}$$

has determinant 1, so it is invertible modulo K. In the finite group of invertible 2×2 matrices modulo K, some positive power is the identity. For such an exponent d put

$$U = \chi_A(d), \qquad V = \psi_A(d).$$

Then $U \equiv 1 \pmod{K}$, $K \mid V$, $V > 0$, and $U \geq A \geq 2$. Set $u = U - 2$ and $v = V/K$. In particular $\gcd(U, 4P_*) = 1$.

Stage 3: the common CRT base. Choose, by CRT and addition of a positive period, a number

$$B > A, \qquad B \equiv A \pmod{U}, \qquad B \equiv 1 \pmod{4P_*}.$$

Take it large enough that $b = (B-1)/(4P_*) - 1 \geq 0$. For each exponent put $S_j = \chi_B(e_j)$ and $R_j = \psi_B(e_j)$. The recurrence modulo U gives $S_j \equiv X_j \pmod{U}$; the recurrence at base 1 modulo $4Y_j$ gives $R_j \equiv e_j \pmod{4Y_j}$. Thus the required integer s_j, r_j exist. The local and global norms hold by construction.

Finally, the Pell power congruence gives

$$X_j + Y_j n_i - Y_j A - n_i^{k_i} \equiv 0 \pmod{T_i}.$$

It follows either directly by expanding powers in the quadratic ring, or from `x_sub_y_dvd_pow` in the supplied arithmetic source. The positive integer T_i therefore supplies λ_i. This proves completeness and Theorem 4.1.

Remark 4.2 (Why the argument can share all calls). There is no circular witness bound. The common bootstrap is chosen after the true input/output tuple. The product of ordinates is chosen after the bootstrap. The auxiliary Pell solution is chosen after that product, and the CRT base after the auxiliary solution. None of these private choices changes a previously fixed input/output value.

5 Sharing neighboring exponents

The actual arithmetic source has forty distinct power calls but only nineteen distinct exponent expressions. Three are immediate shifts of two others:

Derived exponent	Seed exponent	Shift
$L+2$	L	2
$N+2$	N	2
$N+1$	N	1

Here L is the actual factorial scale argument and N the positive proof-row count. The emitted builder proves these identities by exact affine normalization over nonlinear arithmetic nodes. It does not rely on fixed node numbers or an assumed semantic equality.

Let $(X_0, Y_0) = (\chi_A(k), \psi_A(k))$ and $D = A^2 - 1$. Compute

$$X_1 = AX_0 + DY_0, \qquad Y_1 = X_0 + AY_0,$$
$$X_2 = 2AX_1 - X_0, \qquad Y_2 = 2AY_1 - Y_0.$$

These are exactly the pairs at $k+1$ and $k+2$. The second-order expressions use fewer multiplications than a second independent first-order step. They also equal the nonnegative formulas

$$X_2 = (2A^2 - 1)X_0 + 2A(A^2 - 1)Y_0,$$
$$Y_2 = 2AX_0 + (2A^2 - 1)Y_0.$$

Thus the signed intermediate registers do not create additional semantic cases.

Only sixteen seed pairs need Lemma 3.1. Define P_* as the product of just their ordinates. Once the seed pairs are identified, the recurrences identify all derived pairs. The power criterion then applies to every call. No divisibility or index certificate is needed for a pair already defined by an exact recurrence.

Before applying the signed index refinement, this removes twelve semantic inputs and 36 domain witnesses, plus a net 25 nonlinear registers: 73 integer variables, taking 2,042 to 1,969. The signed refinement then removes 96 more coordinates, taking 1,969 to 1,873. These counts state the order of comparison; the recurrence theorem itself applies equally to the signed certificate.

6 The fixed logical input: a direct numerical calculus

The source is the fixed C222 code calculus from ProveIt [6]. The new equation does not enumerate a Turing-machine execution. The relevant inherited interface is the machine-independent theorem `KEndpoint.c222_iff_zf_refutation`, composed with `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Both theorem statements and their source provenance accompany this work. The constructible-model argument is the established logical calibration, not a new consequence of the arithmetic batching lemma.

For clarity the numerical constructors can be printed compactly. Put

$$\mathsf{I}(a,b) = 2(a+b)(a+b+1) + 4b, \qquad \mathsf{E}(a,b) = \mathsf{I}(a,b) + 1,$$
$$\mathsf{Q}(a,b) = \mathsf{I}(a,b) + 2, \qquad \mathsf{J}(a,b) = \mathsf{I}(a,b) + 3,$$
$$\mathsf{N}(a) = \mathsf{I}(a,1), \qquad \mathsf{Ex}(a,b) = \mathsf{N}(\mathsf{Q}(a,\mathsf{N}(b))).$$

Code 0 is truth and code 1 is falsity. The eight ordinary axiom conclusions are

$$\begin{aligned}
\rho_1(a,b) &= \mathsf{Ex}(a,\mathsf{J}(a,b)),\\
\rho_2(a,b,c) &= \mathsf{I}(\mathsf{Q}(a,\mathsf{I}(b,c)),\mathsf{I}(\mathsf{Q}(a,b),\mathsf{Q}(a,c))),\\
\rho_3(a,b,c) &= \mathsf{I}(\mathsf{J}(a,b),\mathsf{I}(\mathsf{E}(c,a),\mathsf{E}(c,b))),\\
\rho_4(a,b,c) &= \mathsf{I}(\mathsf{Q}(a,\mathsf{Q}(b,c)),\mathsf{Q}(b,\mathsf{Q}(a,c))),\\
\rho_5(a,b,c) &= \mathsf{I}(\mathsf{J}(c,a),\mathsf{I}(\mathsf{E}(c,b),\mathsf{E}(a,b))),\\
\rho_6(a,b) &= \mathsf{I}(\mathsf{Ex}(a,\mathsf{Q}(a,b)),b),\\
G(a,b,c) &= \mathsf{I}(\mathsf{J}(c,a),\mathsf{I}(\mathsf{J}(c,b),\mathsf{Q}(\mathsf{E}(a,b),\mathsf{J}(a,b)))),\\
\rho_8(a,b,c,d) &= \mathsf{I}(\mathsf{I}(\mathsf{I}(c,d),a),\mathsf{I}(\mathsf{I}(a,c),\mathsf{I}(b,c))).
\end{aligned}$$

The remaining axiom conclusion is the seed bundle. With

$$\mathrm{req}(b,k,m) = \mathsf{Q}(b,\mathsf{I}(m,\mathsf{N}(\mathsf{Q}(b,\mathsf{I}(m,\mathsf{N}(\mathsf{E}(b,k))))))).$$

To keep the closure constructor readable, introduce three exact abbreviations:

$$\begin{aligned}
\mathrm{sb}(a,b,k) &= \mathsf{Q}(k,\mathsf{I}(\mathsf{E}(k,a),\mathsf{E}(k,b))),\\
\mathrm{c}_1(a,b,k,m) &= \mathsf{Q}(a,\mathsf{I}(\mathrm{sb}(a,b,k),\mathrm{req}(b,k,m))),\\
\mathrm{c}_2(a,b,k,m) &= \mathsf{Q}(a,\mathsf{I}(m,\mathsf{N}(\mathrm{c}_1(a,b,k,m)))),\\
\mathrm{cl}(a,b,k,m) &= \mathsf{Q}(b,\mathsf{I}(\mathsf{E}(b,k),\mathsf{N}(\mathrm{c}_2(a,b,k,m)))).
\end{aligned}$$

define

$$\begin{aligned}
\mathrm{nks}(a,b,c) &= \mathsf{Q}(c,\mathsf{I}(\mathrm{cl}(a,b,c,c),\mathsf{N}(\mathrm{req}(b,c,c)))),\\
\mathrm{vr}(a,b,c) &= \mathsf{I}(\mathsf{E}(a,b),\mathsf{Q}(c,\mathsf{E}(a,b))),\\
\mathrm{bundle}(a,b,c) &= \mathsf{N}(\mathsf{I}(\mathrm{vr}(a,b,c),\mathrm{nks}(a,b,c))).
\end{aligned}$$

Its guard is $c \neq a, b$. The displayed constructors and the literal JSON constructor graph have been independently checked against the pinned code definitions.

There are twelve single-conclusion rule branches: truth; these nine axiom conclusions; modus ponens with premises $a, \mathsf{I}(a,b)$ and conclusion b; and generalization with premise b and conclusion $\mathsf{Q}(a,b)$. Unused premises are code 0. The bundle guard has the exact existential natural form

$$((c-a)(c-b))^2 = (d+1)^2.$$

It holds for some $d \in \mathbb{N}$ exactly when $c \neq a$ and $c \neq b$. All conclusion constructors are polynomials with nonnegative coefficients.

The calibration we inherit states

$$\mathrm{C222}(1) \quad \longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}\,. \tag{17}$$

This is the only logical-strength interface needed by the new direct compiler. The arithmetic proof does not require a semantic interpretation for every possible intermediate natural code.

7 From a finite derivation to eight bounded columns

We summarize the inherited eight-column projection in sufficient detail to identify every proof obligation [4]. The complete source proof is retained in `proofs/SOURCE_PROJECTION.md`.

7.1 Padding and a literal label majorant

For each of the twelve branches let p_b, q_b, v_b be its two premise labels and conclusion after substituting its four natural parameters a, b, c, d. Choose a proof-tree expansion deriving code 1. Pad its list of rule instances with truth steps until its length $N \geq 1$ bounds all four parameters of every step. Define the explicit nonnegative polynomial

$$H(N) = \sum_b \big(p_b(N,N,N,N) + q_b(N,N,N,N) + v_b(N,N,N,N)\big).$$

It bounds every label in the list. Put

$$M_0 = 2^{2H(N)N+1}, \qquad M = M_0 + 1.$$

This power is one of the forty calls later replaced by the simultaneous polynomial certificate.

Use an accumulator with $S_0 = M + 1$, $S_N = 1$, and updates

$$S_{i+1}(1 + v_i M) = S_i(1 + p_i M)(1 + q_i M). \tag{18}$$

For a proof tree, S_i is the product of the factors for outstanding nonzero obligations. Expanding one obligation gives the displayed update; the empty product at the end is 1.

Conversely, all factors are positive and $S_0 > 0$, so induction gives $S_i > 0$ for every i. Telescoping and cancellation in the updates and endpoints therefore imply

$$\prod_i (1 + v_i M) = (1 + M) \prod_i (1 + p_i M)(1 + q_i M).$$

The coefficient magnitudes of the difference of these two polynomials in a formal argument are at most $M_0 < M$. This follows by evaluating the nonnegative coefficient sums at 1 and using $H + 1 \leq 2^H$. A least nonzero coefficient cannot vanish on evaluation at M: reduction modulo M would make it divisible by M. Thus the two polynomials are identical.

Comparing multiplicities at the distinct roots $-1/j$ for positive labels j gives the Horn balance equation

$$\sum_i (e_{v_i} - e_{p_i} - e_{q_i}) = e_1, \qquad e_0 = 0.$$

Let D be the least closure of the selected rules with truth available. If 1 were outside D, the sum of balance coordinates outside D would be nonpositive on each rule: an outside conclusion must have an outside premise. Its value on e_1 is 1, a contradiction. Therefore code 1 is derivable. This proof allows cycles, repeated premises and self-loops.

7.2 The two adjacent beta digits

Encode the finite accumulator by the ordinary remainder relation

$$x = q_i(1 + (i + 1)y) + S_i.$$

Choose y above every S_i and divisible by $N!$; the moduli for $0 \leq i \leq N$ are pairwise coprime, so CRT supplies x. Let U_0 bound all the finitely many quotients. At row i the only bounded local coordinates are

$$(a, b, c, d, S_i, q_i, S_{i+1}, q_{i+1}) \quad \text{with bounds} \quad (N, N, N, N, y, U_0, y, U_0).$$

The bound $S_i \le y < 1 + (i+1)y$ already proves the strict remainder condition. No separate local remainder-bound slack is needed.

For each branch form the square of its accumulator residual and, for the bundle branch, add the square of its guard residual. Multiply all twelve branch sums, then add the squares of the two beta reconstruction residuals. Call the resulting fixed integer polynomial $R(i, \mathbf{w})$. Its zero set is exactly the row relation. The truth, modus ponens and generalization rows use the literal premise/conclusion substitutions above. No independent constructor coordinates cross the bounded quantifier.

8 Signed row evaluation and the absolute majorant

Represent R by a fixed arithmetic graph with operations $+, -, \times$. Its absolute majorant R^+ replaces every subtraction by addition and every literal constant by its absolute value. It is a polynomial graph with nonnegative coefficients.

Lemma 8.1 (Circuit majorant)**.** *If $0 \le i \le N$ and $0 \le w_j \le u_j$, while the other natural parameters are held fixed, then*

$$|R(i, \mathbf{w})| \le R^+(N, \mathbf{u}).$$

Proof. Induct through the graph, using the triangle inequality at addition and subtraction, multiplicativity of absolute value at multiplication, and monotonicity of nonnegative arithmetic at the input bounds. □

We evaluate the signed graph directly for the row congruence and use only its absolute majorant for the range argument. Expanding both halves of a positive/negative pair representation at both interfaces was unnecessary. This saves 80 final integer coordinates compared with the earlier paired evaluation in the synchronized construction. The finite source checker proves the complete signed-row identity compositionally.

9 A bounded-family certificate with lifted CRT periods

Fix the outer parameters and the bounds $\mathbf{u}$. Set

$$L = R^+(N, \mathbf{u}) + N, \quad T = L!, \quad d_i = 1 + (i+1)T, \quad C = \prod_{i<N} d_i.$$

The source has $N \ge 1$, hence $L \ge 1$. The following finite arithmetic conditions express the whole bounded family:

$$C \mid \binom{A_j}{u_j + 1} \ (1 \le j \le 8), \qquad C \mid 1 + (c+1)T, \qquad R(c, \mathbf{A}) \equiv 0 \pmod{C}. \tag{19}$$

The encoded columns are parameterized as $A_j = \alpha_j + 1$, allowing zero decoded values while making the arithmetic upper arguments positive.

Proposition 9.1 (Bounded row equivalence)**.** *Conditions (19) have natural encoded columns and index if and only if every $i < N$ has a natural row witness bounded by $\mathbf{u}$ with $R(i, \mathbf{w}) = 0$.*

Soundness. Choose a prime $p \mid d_i$. Since $\gcd(d_i, T) = 1$, it does not divide $L!$, so $p > L$. The index condition gives $c \equiv i \pmod{p}$. Multiplying a range divisibility by $(u_j + 1)!$ and using the integer falling-factorial identity gives

$$p \mid \prod_{t=0}^{u_j} (A_j - t).$$

Some $w_j \in [0, u_j]$ therefore satisfies $A_j \equiv w_j \pmod{p}$. The row congruence yields $R(i, \mathbf{w}) \equiv 0 \pmod{p}$. The majorant bounds its absolute value by $L < p$, so it is zero. There is no cancellation of a possibly noninvertible factorial here. □

Completeness. The d_i are pairwise coprime: a common prime would divide a nonzero difference of indices smaller than N and hence would divide $L!$, which contradicts $d_i \equiv 1 \pmod{L!}$. For each column put $F_j = (u_j + 1)!$ and lift the periods to

$$D_{ij} = \prod_{p \mid d_i} p^{v_p(d_i) + v_p(F_j)}.$$

For fixed j these moduli are still pairwise coprime. Choose $A_j \equiv w_{ij} \pmod{D_{ij}}$ for all i, and shift by their product to make $A_j > u_j$. The numerator of the falling factorial has a factor with enough valuation to absorb both d_i and F_j. Division by F_j therefore leaves $d_i \mid \binom{A_j}{u_j+1}$. Choose the encoded index independently modulo the original d_i. The strengthened column congruences imply the original row congruences. All of (19) follows. □

The lifted periods occur only in the proof of existence; they add no variable or prime-factorization operator to the emitted polynomial. A shift by the unlifted product is insufficient: for example $15 \mid \binom{0}{5}$ but $15 \nmid \binom{15}{5} = 3003$. This is why merely requesting positive encoded columns without a compatible-period proof would be invalid.

10 Eliminating the remaining arithmetic operations

The CRT product has the explicit formula

$$Q = (1 + (N+1)T)^N + 2,$$
$$C = \left(T^N N! \binom{Q+N}{N}\right) \bmod (TQ - 1).$$

Modulo $TQ - 1$, the numerator is congruent to $\prod_{j=1}^{N}(1 + jT)$ because $TQ \equiv 1$. Also that product is at most $(1 + NT)^N < Q - 1 \le TQ - 1$, so the remainder is exactly the desired positive product.

For a binomial call with positive n, k, let $q = 2^n + 1$, $D = q^k$ and $F = (q+1)^n$. Use

$$F = D(qh + b) + r, \qquad D = r + r_s + 1, \qquad q = b + b_s + 1,$$

with $b, r, r_s, b_s \in \mathbb{N}$ and $h \in \mathbb{Z}$. If $h < 0$, then $qh + b < 0$ and the first equation would give $F < 0$. Thus $h \ge 0$ is forced. The displayed relations say that b is the base-q digit of $\lfloor F/D \rfloor$. Every binomial coefficient of order n is smaller than q, so $b = \binom{n}{k}$, including the case $k > n$.

For a factorial argument $n \ge 1$, put

$$R_0 = (n+2)^{n+2}, \qquad c = \binom{R_0 + n}{n}.$$

The explicit factorial identity in the supplied source is

$$n! = \left\lfloor \frac{(R_0+1)^n + c - 1}{c} \right\rfloor.$$

The quadratic-graph interface imposes

$$(R_0+1)^n + c = cf + r + 1, \qquad c = r + s + 1,$$

with natural remainder and slack. We parameterize $f = f_0 + 1$. This is valid because every factorial is positive.

These constructions leave forty power calls, eleven binomial calls, two factorial calls and three ordinary division calls. Each power output is parameterized as $m = m_0 + 1$. The actual graph then establishes *all forty base and exponent positivity conditions syntactically*, before assuming any power or factorial equality. This avoids a circular application of the simultaneous theorem.

The nine coordinate/index divisibility quotients are unrestricted: their natural numerators and the now-proved positive CRT product force them to be nonnegative. Similarly the three ordinary Euclidean quotients are forced nonnegative by a natural dividend and a bounded natural remainder. These facts justify the omitted guards. The checkers verify that these inferred signs do not feed a power-argument positivity proof.

11 From the mixed arithmetic graph to a quartic over integers

For every semantic input x still required to be nonnegative, add three fresh unrestricted integers a, b, c and the quadratic residual

$$x - a^2 - b^2 - c(c+1) = 0. \tag{20}$$

The right side of $x = a^2 + b^2 + c(c+1)$ is always nonnegative. Conversely, Legendre's three-square theorem applies to $4x + 1$ for every $x \geq 0$: it is not of the form $4^a(8b+7)$. Modulo four, precisely one of the three representing squares is odd. Writing the even squares as $(2a)^2, (2b)^2$ and the odd one as $(2c+1)^2$ yields (20).

All other arithmetic registers range freely over $\mathbb{Z}$. Substitute addition, subtraction and scalar multiplication as affine forms. For every genuine product introduce a fresh variable z and the quadratic equation $z - L_1 L_2 = 0$, where L_1, L_2 are the current affine forms. If an affine form exceeds sixteen monomials, retain it as a register with a linear defining equation. The smaller-height variant uses a threshold of one.

Given semantic coordinates, topological evaluation supplies unique integer values for every retained register. Conversely, the defining equations force those values in order. Hence this flattening preserves the complete integer zero set up to unique register restoration. The nonnegativity representations are existentially surjective, as proved above. Finally a sum of integer squares vanishes exactly when all its summands vanish.

Component	Variables	Residuals
Semantic inputs	245	–
Source arithmetic equalities	–	57
Shared power-certificate equalities	–	75
Nonlinear defining registers	1,153	1,153
Retained affine registers	37	37
146 nonnegativity representations	438	146
Total	**1,873**	**1,468**

Of the 245 semantic inputs, 146 require nonnegativity and 99 are unrestricted. The latter include the 32 newly unguarded Pell progression coordinates. The degree is at most four by construction, and exactly four because a fresh guard coordinate has fourth-power coefficient 1 and occurs in no other residual. The collected expansion independently confirms the degree, every coefficient and all-variable liveness.

12 Proof of the consistency theorem

Every step in the following chain has been given both existence directions:

$$\begin{aligned} P(\mathbf{z}) = 0 \text{ for some } \mathbf{z} \in \mathbb{Z}^{1873} &\iff \text{the mixed arithmetic certificate has a solution} \\ &\iff \text{the bounded family of direct proof rows holds} \\ &\iff \mathrm{C222}(1) \\ &\iff \mathrm{Ref}_{\mathrm{ZFC}}\,. \end{aligned}$$

The first equivalence uses the register restoration and (20). The second uses the signed shared Pell theorem, recurrence substitution, exact binomial/factorial/division graphs and the lifted CRT theorem. The third is the padded Horn-flow construction. The last is the pinned logical calibration (17).

All new arguments are finite algebra, elementary integer arithmetic, Pell recurrence theory, finite-group reasoning, CRT and finite proof combinatorics. They can be carried out inside ZFC over its own internal ω. Neither direction is a meta-level appeal to the truth of consistency. The chain proves Theorem 1.1; classical negation gives its no-zero/Con(ZFC) consequence.

The attached Lean files document exact inherited theorem interfaces. They are not advertised as a standalone compilable Lean project, and we have not emitted an internal proof object for the new composed biconditional. This distinction does not turn the ordinary argument into a conjecture; it states the form of the evidence being delivered.

13 Reproduction, exact checks and provenance

All Python programs use only the standard library. From the package root:

```
python3 build.py
python3 build.py --affine-cap 1 --output equations_small_height
python3 verify_release.py
```

The release verifier reconstructs the two equations in temporary directories, compares every generated byte, and runs the independent checks in normal and optimized Python. The mathematical proofs remain necessary: evaluating finitely many integer tuples could not establish the infinite zero-set equivalence.

The independent checks have complementary scopes:

- `verify_source.py`: every one of the twelve rule branches, the paired and signed local row, the absolute majorant, the full heterogeneous bounds, both beta endpoints, all 57 source equalities, and every arithmetic-call interface. It checks unconditional positivity of all forty power bases and exponents. Its companion `verify_source_negative.py` rejects three corruptions of the bounds, majorant and horizon domain.

- `verify_batch.py`: all common definitions, seed and derived Pell coordinates, every common/local norm, all forty power congruences, the exact domain choices and the entire 75-row batch. Its negative controls corrupt meaningful formulas and required domain guards.
- `verify_recurrence.py`: the free polynomial recurrence identities, the three actual exponent differences and the exact product of sixteen seed ordinates.
- `verify_polynomial.py`: complete pruning, every affine and nonlinear register equation, all nonnegativity rows, and exact collection of every coefficient in the sum of squares. A coefficient change and a guard-square change are detected.

The checkers do not import the equation builder. The coefficient checker uses symmetric convolution, independently of the emitter's ordered double multiplication loop. Checks use explicit exceptions rather than assertions that disappear under `python3 -O`.

13.1 The finite encoding of the equation

The polynomial JSON uses a list of pairs `[coefficient,monomial]`. A monomial is a sorted list of zero-based variable indices, with repeats representing powers. For example `[7,[0,0,3]]` means $7z_0^2z_3$, and the empty list denotes 1. Coefficients are decimal integer literals. The residual JSON uses the same convention. These rules specify the entire equation without requiring the reader to execute the generator.

The primary polynomial's SHA-256 is

```
18d09e54a3d66ed33422560a35724a5073
fd1ca1a2a76bfaf16e634cb019ea1d
```

The line break is typographical; concatenate the two lines. The release checksum manifest covers the immutable source, equation, proof and document artifacts. The regenerable `checks/` receipts and the manifest itself are excluded.

13.2 Attribution and proof boundary

The eight-column source, Horn-flow argument, C222 calibration, prime-power CRT lift, merged binomial/factorial formulas and three-square integer conversion have explicit predecessors in the inspected repository. The new contributions in this package are their concrete synthesis with the shared bootstrap/CRT power certificate, the one-gap output bound, the signed positive-index recovery theorem, exact neighboring-exponent sharing, the absolute-majorant evaluation and the resulting fully materialized equations. The basic Pell theorems and CRT are classical.

The report makes no claim of optimal degree/variable/height tradeoff, historical uniqueness, short consistency proofs, or small possible refutation witnesses. It supplies a fixed equation, an ordinary proof of its exact target semantics, and reproducible finite artifacts.

14 What this changes for the next iteration

The main structural gain is a compatible common certificate for an entire finite family of powers. The construction uses two global Pell norms and two local norms per seed exponent. The theorem is reusable with a different logical source; its proof states how all shared witnesses can be chosen.

References

[1] Vladimir Reshetnikov and ProveIt contributors, *ProveIt*, inspected commit `b594a5e988e986f695923d068ef3d33ab2ada16f`. https://github.com/VladimirReshetnikov/ProveIt/tree/b594a5e988e986f695923d068ef3d33ab2ada16f/Computability/DiophantineConsistency.

[2] ProveIt, *Later arrivals and current size reductions*, `Computability/DiophantineConsistency/PROPOSAL_REVIEW_LATEST.md`, at the inspected commit. Includes the direct C222 and shared CRT comparison points and the ordinary/formal proof boundaries.

[3] ProveIt, *Active equation-size research*, `Computability/DiophantineConsistency/SIZE_REDUCTION.md`, at the inspected commit. Its table distinguishes domains, degree, operation count and the native-coefficient quartic alternatives.

[4] *ConZFC Direct Projection 7617*, incoming archive `docs/incoming/ConZFC_Direct_Projection_7617.zip`, at the inspected commit. Its literal rule graph, source projection proof, arithmetic interfaces and logical pins are retained or explicitly transformed in this package.

[5] ProveIt, *Smaller direct quartics from a prime-power CRT lift*, `candidates/shared-crt-prime-lift/PROOF.md`, under the inspected DiophantineConsistency directory.

[6] ProveIt, `KCode222.lean`, `KEndpoint.lean` and `ConstructibleChoice.lean`, logical source pin `9a33342a03d19cf5b061b2083c6b5d3f27f752ed`. The full filenames, URLs and SHA-256 values are in `provenance/LOGICAL_SOURCE_PINS.json`.

[7] Mario Carneiro and the Mathlib contributors, *Pell's equation and Matiyasevic's theorem*, `Mathlib/NumberTheory/PellMatiyasevic.lean`. https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/PellMatiyasevic.html. The exact source blob used is copied in `provenance/`, with its Apache 2.0 license.